

Chapter 1

THE PROBLEM AND ITS BACKGROUND

Introduction

Disease surveillance is defined as the systematic and continuous collection, analysis, interpretation, and dissemination of health data essential for planning, implementing, and evaluating public health practice (World Health Organization [WHO], 2018). Effective surveillance systems enable health authorities to detect outbreaks early, guide targeted interventions, and allocate resources based on spatial and temporal patterns of disease occurrence. Globally, the integration of artificial intelligence (AI) and machine learning into public health surveillance has significantly expanded the analytical capabilities available to health agencies, enabling automated outbreak detection, predictive forecasting, and data-driven decision support that traditional manual methods cannot provide (Srivastava et al., 2024).

In the Philippines, the Department of Health (DOH) implements the Philippine Integrated Disease Surveillance and Response (PIDSR) system as the national framework for monitoring notifiable diseases (DOH, 2014). While PIDSR provides a standardized reporting structure through line listings and tabular aggregation, its implementation at the city epidemiology level remains largely spreadsheet-based and limited in spatial visualization and automated outbreak detection capability. The Centers for Disease Control and Prevention (CDC, 2019) has emphasized that electronic and digital surveillance systems significantly improve timeliness, completeness, and response efficiency, highlighting the operational gap that persists in many local government health offices.

Parañaque City, as one of the highly urbanized cities in Metro Manila, Philippines, generates substantial weekly disease data through its network of health centers, hospitals, and sentinel sites. The City Epidemiology Section Unit (CESU) is responsible for monitoring disease trends and recommending public health responses. Despite maintaining disease records traceable to 2007 under the PIDS framework, the CESU currently processes surveillance data using manual spreadsheet tools that offer no geospatial visualization, no automated threshold alerting, and no AI-assisted forecasting capability. This limitation reduces the spatial precision and forward-looking orientation of disease monitoring within the city, reflecting the broader challenge identified by Srivastava et al. (2024) that traditional surveillance methods are constrained by manual data collection and slow reporting.

Existing machine learning studies on dengue forecasting in the Philippine context, including Carvajal et al. (2018) in Metropolitan Manila and Olana et al. (2025) at the national level, have demonstrated that AI models such as Random Forest and Prophet can achieve reliable predictive accuracy for disease trend forecasting. However, these studies produced standalone forecasting models without embedding their outputs into operational, web-based surveillance dashboards accessible to city-level government health personnel. This gap between demonstrated AI capability and operational deployment at the local health unit level provides the direct motivation for H-MAP.

The City Government of Parañaque maintains an Employee Portal at hrmo.paranaque.gov.ph, provisioned by the Department of Information and Communications Technology (DICT) and managed by the Human Resource Management Office (HRMO). This existing infrastructure provides a cost-effective foundation for deploying H-MAP as a path-based web application at hrmo.paranaque.gov.ph/hmap. By leveraging the HRMO portal's

existing authentication system, access to sensitive disease surveillance data will be restricted exclusively to active government employees whose employment status is maintained in real time by the Human Resource Management Office, directly addressing the access control requirements of Republic Act No. 10173, the Data Privacy Act of 2012 (Republic Act No. 10173, 2012).

This study proposes the development of H-MAP: A Heat Map Disease Surveillance and Outbreak Prediction System for the City Epidemiology Section Unit of Parañaque City. The proposed system will be implemented as a PHP-based web application integrated with an AI-powered Python machine learning microservice, a MySQL database, Leaflet.js interactive maps, and Chart.js data visualizations, all deployed within the existing DICT-provisioned government server infrastructure of Parañaque City.

Background of the Study

In today's rapidly evolving landscape of public health challenges, the City Government of Parañaque recognizes the critical role of technological advancements in effectively addressing disease outbreaks and safeguarding the health of its residents. The implementation of a comprehensive web-based disease surveillance platform presents a promising solution to enhance disease monitoring and outbreak prediction capabilities. This research project seeks to explore the development of H-MAP, specifically tailored to meet the unique surveillance needs of Parañaque City and deployable within its existing government IT infrastructure without additional licensing costs.

The City Epidemiology Section Unit currently maintains disease records dating back to 2007 in the PIDSR format, covering 28 notifiable diseases across Parañaque City's 16 barangays. This dataset, encompassing over 30,000 patient-level records as of 2023, represents

a significant untapped resource for spatial and temporal disease analysis. Despite its depth, this historical record has not been leveraged for AI-assisted outbreak forecasting or spatial risk visualization. Without a dedicated analytical platform, health officials are unable to generate heat maps, compute epidemic thresholds automatically, or access machine learning-based forecasts from this data. Disease monitoring remains reactive and dependent on manually prepared spreadsheet summaries that do not communicate spatial distribution or trend trajectories.

The main problem identified in this research is the absence of a web-based, AI-integrated disease surveillance system for the City Epidemiology Section Unit of Parañaque City. Without such a system, health officials face constraints in visualizing where diseases are concentrated geographically, identifying barangays at elevated risk, detecting anomalous weekly case spikes in real time, and forecasting near-term disease trajectories based on historical seasonal patterns. These limitations reduce the timeliness and spatial precision of public health interventions, which is a recognized challenge in surveillance systems that rely on passive reporting mechanisms (WHO, 2018; CDC, 2019).

The causes of this gap are multi-layered. Proprietary GIS platforms carry licensing costs that are prohibitive for city-level local government unit health offices operating under fixed annual budgets. The nationally maintained PIDSR electronic reporting system provides case intake and national-level aggregation but does not include city-specific spatial visualization, threshold alerting, or AI-assisted predictive analytics (DOH, 2014). Meanwhile, local studies by Carvajal et al. (2018) and Olana et al. (2025) confirm that machine learning models can reliably forecast dengue trends in the Philippine context but were not integrated into operational platforms accessible to non-technical health personnel at the city level.

In response to the identified problem, H-MAP is proposed as a web-based AI-assisted disease surveillance system built entirely on open-source and government-owned infrastructure. The proposed system will be developed using PHP, consistent with the existing HRMO portal, and will integrate a Python-based machine learning microservice for AI-powered forecasting and barangay risk classification. By deploying H-MAP at hrmo.paranaque.gov.ph/hmap, the study avoids additional server procurement, domain registration, and SSL certificate costs. Authentication will be handled through the HRMO portal's existing session management system, ensuring that only active, HR-verified city employees can access disease surveillance data.

Objectives of the Study

To develop a Web-Based HMAP: A Heat Map Disease Surveillance and Outbreak Prediction System for the City Epidemiology Section Unit of Parañaque City.

General Objective

To develop H-MAP: A Heat Map Disease Surveillance and Outbreak Prediction System for the City Epidemiology Section Unit of Parañaque City, implemented as a web-based application integrated with AI-assisted machine learning capabilities and deployed within the existing DICT-provisioned government server infrastructure.

Specific Objectives

1. To design a comprehensive web-based AI-assisted disease surveillance system with the following features:
 - a. An interactive heat map visualization module using Leaflet.js and OpenStreetMap that will display disease case density across Parañaque City's 16 barangays.

- b. A data digitization and records management module for encoding disease case records consistent with PIDS reporting standards (DOH, 2014).
 - c. A real-time disease trend monitoring module with automated epidemic threshold alerts based on the WHO Early Warning and Response Network (EWARN) methodology (WHO, 2018).
- 2. To develop the H-MAP system using the following technology stack:
 - a. PHP as the primary server-side scripting language, consistent with the existing HRMO portal.
 - b. MySQL as the relational database management system hosted on the DICT-provisioned server.
 - c. Leaflet.js with OpenStreetMap tiles for interactive, license-free web mapping.
 - d. Chart.js for trend visualization and epidemiological dashboard charts.
 - e. Python with Flask as an internal AI machine learning microservice implementing Prophet time-series forecasting and Random Forest barangay risk classification (Olana et al., 2025; Carvajal et al., 2018).
 - f. HRMO portal session-based authentication to restrict access to active Parañaque City government employees, consistent with Republic Act No. 10173 (2012).
- 3. To test and improve the functionality of H-MAP in terms of:
 - a. Functionality to ensure all system modules operate as intended, including authentication, heat map rendering, data encoding, threshold alerting, and AI machine learning prediction outputs.
 - b. Reliability to ensure the system performs consistently under normal operational conditions, including the AI machine learning microservice.

- c. Usability, following ISO 9241 principles, to ensure the system is accessible and intuitive for health officials and data encoders with varying levels of technical expertise.
- 4. To evaluate the performance of H-MAP using the ISO/IEC 25010 software quality model in terms of:
 - a. Functionality, Reliability, Usability, Performance Efficiency, and Maintainability (International Organization for Standardization [ISO], 2011).
 - b. Collecting structured feedback from CESU health officials and IT professionals to identify areas for improvement.

Significance of the Study

The development of H-MAP addresses a concrete and documented operational gap in the disease surveillance capacity of the City Epidemiology Section Unit of Parañaque City. By providing an integrated web-based platform for heat map visualization, automated epidemic threshold monitoring, and AI-assisted outbreak forecasting, this study aims to equip health officials with analytical tools that directly improve the speed, spatial precision, and forward-looking orientation of disease surveillance. The significance of H-MAP is particularly pronounced given that it is designed to be deployed on existing government infrastructure without additional licensing costs, making it a sustainable and replicable model for other city LGUs facing similar surveillance limitations.

The City Epidemiology Section Unit and its health officials are the primary direct beneficiaries of H-MAP. The proposed system will streamline the translation of raw PIDSR case records into actionable geospatial intelligence, reduce the manual workload associated with weekly data compilation and chart preparation, and introduce AI-generated outbreak

alerts that enable earlier detection of anomalous disease trends. Health officers, barangay health workers, and city health decision-makers who currently rely on manual spreadsheet summaries will benefit from a consolidated, map-integrated dashboard that presents disease distribution, AI-assisted trend analysis, and risk classifications in a single interface accessible through any web browser.

The residents of Parañaque City are the ultimate beneficiaries of improved disease surveillance. Earlier outbreak detection, enabled by automated threshold alerts and AI-assisted forecasts, translates into faster public health response, more targeted intervention in high-risk barangays, and more efficient allocation of limited public health resources. In a densely populated urban setting such as Parañaque City, where diseases like Dengue, Measles, and Leptospirosis present recurring seasonal threats, the ability to anticipate and respond to outbreaks proactively rather than reactively has direct implications for reducing morbidity and mortality.

Beyond its immediate operational impact, this study contributes to the applied information technology research literature in Philippine public health. By demonstrating that an open-source, PHP and Python-based architecture can deliver spatial visualization, automated alerting, and AI machine learning forecasting on a government-owned server without proprietary GIS licensing, H-MAP provides a technically documented and evaluated precedent for cost-effective AI-assisted digital health surveillance at the city LGU level in the Philippines. Critically, H-MAP addresses the specific gap identified in the Philippine machine learning literature by Carvajal et al. (2018) and Olana et al. (2025): the absence of an operational, web-based platform that makes AI-generated disease forecasts accessible to non-technical health personnel at the city epidemiology unit level.

Scope and Limitations of the Study

The study covers the design, development, testing, and evaluation of H-MAP: A Heat Map Disease Surveillance and Outbreak Prediction System for the City Epidemiology Section Unit of Parañaque City. The proposed system will be deployed as a path-based web application at hrmo.parañaque.gov.ph/hmap on the existing DICT-provisioned server used by the Parañaque City HRMO portal. The study will make use of historical PIDS disease case data provided by the City Epidemiology Section Unit, covering the years 2010 to 2023 and comprising over 30,000 patient-level records across 16 barangays and multiple notifiable disease categories (DOH, 2014).

The scope of H-MAP includes five functional modules: an HRMO-integrated authentication module; a data digitization and records management module for encoding PIDS-compliant case data; an interactive heat map visualization module built with Leaflet.js; a disease trend monitoring module with automated WHO EWARN epidemic threshold alerting (WHO, 2018); and an AI machine learning prediction module implemented as an internal Python Flask microservice delivering Prophet-based weekly case forecasts and Random Forest barangay risk classifications, consistent with methodologies validated in the Philippine context by Carvajal et al. (2018) and Olana et al. (2025).

The study does not include actual public health intervention planning, clinical patient management, or biometric data processing. The AI machine learning outputs of H-MAP are intended as decision-support indicators and do not replace the professional judgment of trained epidemiologists. The system is limited to the 16 barangays of Parañaque City and does not include cases from adjacent cities recorded in the dataset. Evaluation will be conducted using an ISO/IEC 25010-based questionnaire administered to purposively selected health official and

IT professional respondents within a defined evaluation period (ISO, 2011). The study does not include long-term system deployment monitoring, high-concurrency stress testing, or integration with the national DOH PIDSR electronic reporting portal.

Chapter 2

CONCEPTUAL FRAMEWORK

Review of Related Literatures and Studies

This chapter presents a review of related literature and studies that provide the theoretical and empirical foundations of the proposed H-MAP system. Conceptual literature is organized around the key technical and public health domains that the system addresses. Related research studies and systems, both local and international, are reviewed and synthesized to identify the research gap that the proposed development aims to address. The chapter concludes with the Input-Process-Output (IPO) conceptual framework and the operational definition of terms.

Disease Surveillance and the PIDSR Framework

Disease surveillance is defined as the systematic and continuous collection, analysis, interpretation, and dissemination of health data essential for planning, implementing, and evaluating public health practice (WHO, 2018). Surveillance systems enable early detection of outbreaks and guide public health interventions. The WHO (2018) identifies disease surveillance as a core function of national health systems and emphasizes that strengthening surveillance infrastructure requires both improved reporting accuracy and enhanced analytic capacity.

Surveillance systems are commonly classified into four types: passive, active, sentinel, and syndromic. Passive surveillance depends on routine reporting from health facilities, while active surveillance involves proactive case identification by health authorities. Sentinel surveillance monitors selected reporting sites that serve as representative indicators of disease trends, and syndromic surveillance detects patterns based on symptoms before laboratory

confirmation is obtained (WHO, 2018). Although passive surveillance remains widely used in developing countries including the Philippines, limitations in timeliness and spatial interpretation persist as recognized challenges. The CDC (2019) has emphasized that enhancing passive systems with digital data management and geospatial visualization improves detection of outbreak clusters and transmission patterns.

In the Philippines, the DOH implements the PIDSR system as the national disease notification and reporting framework. Under PIDSR, health facilities are required to report cases of 28 priority diseases using standardized case report forms that include disease classification, case outcome, patient demographics, reporting facility, and barangay-level geographic location (DOH, 2014). While PIDSR ensures uniform data collection, the system does not inherently provide advanced geospatial analytics or automated statistical outbreak detection at the city epidemiology level. H-MAP is designed as a supplementary analytical layer on top of existing PIDSR data, enhancing the local surveillance capabilities of the CESU without replacing or duplicating the national reporting workflow.

Geographic Information Systems in Public Health

Geographic Information Systems are computer-based platforms designed to capture, store, analyze, and visualize geographically referenced data. In public health, GIS has become a powerful tool for mapping disease distribution and identifying spatial relationships among cases (Nykiforuk & Flaman, 2011). Nykiforuk and Flaman (2011) conducted a comprehensive review of GIS applications in health promotion and found that GIS-based approaches consistently provided superior capacity for spatial disease visualization compared to traditional tabular analysis, enabling more targeted and efficient public health interventions. Lawson (2018) similarly explains that spatial epidemiology uses GIS tools to detect clustering patterns

and assess environmental risk factors, transforming raw epidemiological data into actionable intelligence that enables faster and more informed decisions by health authorities.

The emergence of open-source and web-based GIS technologies has substantially reduced the barrier to adoption for local health agencies. Boulos et al. (2011) reviewed web-based GIS technologies in public health and found that open-source web mapping platforms provide sufficient functionality for disease mapping tasks previously requiring proprietary desktop GIS software, with ease of deployment, cross-platform accessibility, and zero licensing costs identified as key practical advantages for health agencies in resource-constrained settings. Leaflet.js, the open-source JavaScript library for interactive web mapping selected for H-MAP, renders maps directly in the browser without requiring client-side software installation, which is a practical advantage in health office environments where workstation configurations vary.

Geographic base map data for H-MAP will be sourced from OpenStreetMap, and barangay boundary data will be derived from the Philippine GIS Data Exchange Center (PhilGIS) under an open data license. This approach avoids proprietary licensing costs while delivering the spatial reference framework needed for barangay-level disease case geocoding and heat map generation, consistent with DICT's open-source government systems policy.

Heat Mapping and Spatial Cluster Detection

Heat maps represent disease case density using color gradients to visualize intensity distribution across a geographic area. Areas with high concentrations of cases appear in warmer colors, while lower concentrations appear cooler, enhancing situational awareness by enabling immediate recognition of outbreak hotspots (Lawson, 2018). Spatial visualization of

this type enhances public health planning while minimizing individual-level exposure of patient information, particularly when aggregated to the barangay or district level.

Kernel Density Estimation (KDE) is the primary spatial statistical method underlying heat map generation. KDE calculates a smoothed estimate of case density at each point in a geographic space based on the distribution of reported cases, producing a continuous, visually intuitive surface of disease concentration (Lawson, 2018). Spatial autocorrelation methods complement KDE by identifying whether the clustering of cases is statistically significant beyond what would be expected by chance. These cluster detection methods strengthen outbreak detection by revealing patterns that may not be evident in tabular data, thereby supporting earlier intervention strategies (Lawson, 2018).

Dong et al. (2020) demonstrated the wide-scale operational effectiveness of spatial visualization during the COVID-19 pandemic. Their development of the Johns Hopkins University COVID-19 Dashboard, which integrated real-time GIS mapping with case count data, showed that geospatial visualization in a web-based format significantly improved global situational awareness and informed public health decision-making. This system serves as an internationally recognized benchmark for the dashboard design approach adopted in H-MAP, particularly the integration of map-based visualization with tabular and chart-based data summaries.

Epidemic Threshold Methods and Automated Outbreak Detection

Outbreak detection mechanisms in epidemiological surveillance commonly rely on statistical aberration detection methods that compare current case counts against historical baselines. The most widely used approach, consistent with the WHO Early Warning and Response Network (EWARN) methodology, defines the epidemic threshold as the mean

number of cases recorded during the same epidemiological week across the previous five years, plus two standard deviations (WHO, 2018). When current case counts exceed this value, an outbreak warning is generated. This can be expressed as:

$$\text{Alert if: Current Cases} > \text{Baseline Mean} + 2 (\text{Standard Deviation})$$

This threshold model is epidemiologically principled because it accounts for both the expected seasonal level of disease activity and the year-to-year variability in case counts. The same mathematical basis is used in the Philippines' own national PIDSR five-year average alert tables, prepared annually by the National Epidemiology Center (DOH, 2014). The CDC developed the Early Aberration Reporting System (EARS), which utilizes moving averages and standard deviation thresholds to identify abnormal disease activity (CDC, 2019). Time-series moving averages smooth temporal fluctuations in case counts and detect aberrations, expressed as:

$$\text{MA}(t) = [C(t-1) + C(t-2) + C(t-3)] / 3$$

Where $C(t)$ represents case counts at time t . These threshold approaches are widely accepted in surveillance systems because they balance simplicity, interpretability, and early detection capability (CDC, 2019; WHO, 2018). H-MAP will automate this threshold calculation for all monitored diseases using the 14-year historical PIDSR dataset available from the Parañaque City CESU.

Artificial Intelligence in Disease Surveillance

Artificial intelligence has emerged as a transformative capability in disease surveillance, enabling health systems to move beyond passive data collection toward automated detection, prediction, and decision support. Srivastava et al. (2024) documented the expanding role of AI in early diagnosis and detection of infectious diseases, noting that AI-

assisted surveillance systems analyze historical data patterns, generate automated alerts, and support clinical and public health decision-making in ways that manual surveillance cannot match. Their review emphasized that AI's capacity to identify subtle patterns across large historical datasets makes it particularly valuable in settings where disease data has been collected over many years but has not been systematically analyzed for predictive purposes.

The WHO (2018) and CDC (2019) similarly acknowledge that digital and AI-enabled surveillance systems significantly improve timeliness and detection accuracy over traditional methods. The WHO (2018) highlights that early warning systems incorporating automated analytical models reduce the time between outbreak onset and health authority notification, enabling earlier and more effective public health response. The CDC (2019) has actively developed AI-assisted surveillance tools including the Early Aberration Reporting System, which operationalizes statistical learning principles for automated anomaly detection in disease incidence data, demonstrating the practical value of embedding AI within routine surveillance workflows.

For H-MAP, AI is embedded in two specific components: the Prophet time-series forecasting model, which applies additive regression and automatic seasonality detection to generate four-week-ahead disease case predictions, and the Random Forest classifier, which uses ensemble learning to classify each of Parañaque City's 16 barangays by disease risk level for a given week. Together, these components constitute H-MAP's AI-assisted early warning capability, providing health officials with forward-looking intelligence that complements the automated epidemic threshold alerts. This design reflects the principle identified by Srivastava et al. (2024) that AI in surveillance should serve as a decision-support layer rather than a replacement for epidemiological expertise.

Machine Learning for Disease Forecasting: Prophet Time-Series Models

Predictive modeling applies statistical and time-series techniques to forecast disease trends using historical incidence data. Shaman and Karspeck (2012) demonstrated that influenza outbreaks can be forecasted using time-series modeling and statistical inference techniques, establishing the foundational evidence that machine learning-based forecasting can provide operationally useful lead time for public health preparedness. Prophet, developed by Meta's Core Data Science team and released as open-source software, decomposes time-series data into trend, yearly seasonality, weekly seasonality, and holiday effect components, fitting these through an additive regression model that is robust to missing data and irregular reporting patterns (Taylor & Letham, 2018).

Olana et al. (2025) compared SARIMA, Neural Network Autoregression, Random Forest, LSTM, TBATS, and Prophet for dengue forecasting at the national level in the Philippines using monthly data from 2017 to 2024. Their study confirmed the strong seasonality of Philippine dengue incidence and demonstrated that time-series models including Prophet can produce reliable forecasts when trained on sufficiently long historical data, providing the most current evidence in the Philippine literature for the forecasting approach employed in H-MAP. Chakraborty et al. (2019) similarly applied Prophet to dengue forecasting and reported that the model's automated seasonality handling produced forecasts with substantially lower error rates than traditional ARIMA models, particularly for diseases with strong and regular seasonal patterns. These findings directly validate the Prophet forecasting component of H-MAP, which will target a four-week forecast horizon for Dengue and other priority diseases exhibiting consistent seasonal behavior in the Parañaque City PIDSR dataset.

It is important to note that both Olana et al. (2025) and Chakraborty et al. (2019) produced standalone forecasting models without integration into operational dashboard platforms. H-MAP addresses this limitation by embedding Prophet forecast outputs directly within an authenticated, role-based web dashboard accessible to city epidemiology unit staff without requiring them to operate machine learning tools directly.

Machine Learning for Barangay Risk Classification: Random Forest

Random Forest is an ensemble machine learning method that constructs multiple decision trees during training and outputs the class predicted by the majority of trees (Breiman, 2001). It is valued in classification tasks for its robustness to overfitting, its ability to handle mixed feature types, and its capacity to capture complex interactions between features. In the Philippine context, Carvajal et al. (2018) applied Random Forest, alongside General Additive Modeling, SARIMA, and Gradient Boosting, to forecast dengue incidence in Metropolitan Manila using meteorological and historical case data. Their study found that Random Forest achieved competitive predictive accuracy and identified temporal patterns that single-technique models could not reliably capture, providing direct local evidence for the suitability of Random Forest in Philippine dengue surveillance contexts.

Shi et al. (2016) demonstrated that ensemble learning methods outperformed logistic regression and single decision tree approaches in classifying high-risk geographic units based on historical case patterns and seasonal features in a dengue surveillance context. For H-MAP, the Random Forest classifier will be trained on 14 years of Parañaque City PIDSR data to predict Low, Moderate, or High disease risk for each of the city's 16 barangays for a given week and disease, using features including barangay identity, morbidity week, calendar month, recent case counts, prior-year case counts, and the five-year historical mean. The combination

of epidemic threshold alerting at the city level with Random Forest risk classification at the barangay level provides a layered analytical design that addresses both outbreak detection and geographic targeting of public health resources.

As with the Prophet forecasting component, the Random Forest classifier in H-MAP addresses the operational gap identified in the local literature: Carvajal et al. (2018) produced a machine learning model for Metro Manila dengue forecasting but did not integrate it into a platform usable by health officers in routine surveillance operations. H-MAP converts the classifier's outputs into a color-coded barangay risk overlay on the Leaflet.js map, making AI-generated risk assessments directly interpretable by health personnel without machine learning expertise.

PHP-Based Web Systems and HRMO Portal Integration

PHP is a widely used server-side scripting language specifically designed for web development, supporting dynamic content generation, session management, database connectivity, and form handling. Its adoption in Philippine government web systems is well established, with many LGU portals, eServices platforms, and information management systems implemented in PHP due to its open-source licensing and broad server compatibility. The Parañaque City HRMO portal at hrmo.paranaque.gov.ph is implemented in PHP, making PHP the natural choice for H-MAP to ensure compatibility with the existing codebase, session infrastructure, and database configuration of the host environment.

Session-based authentication in PHP provides a practical mechanism for integrating H-MAP with the HRMO portal's access control system. When an employee logs into the HRMO portal, PHP creates a server-side session storing the employee's identifier, name, role, and employment status. H-MAP will access this session data directly, verifying that the requesting

user is both authenticated in the HRMO portal and recorded as an active employee in the HRMO MySQL database before granting access to disease surveillance features. This approach eliminates the need for a separate H-MAP user database, reduces administrative overhead, and ensures that access control inherits the HRMO office's real-time employee status records, including immediate revocation of access upon HR recording of an employee separation. This design directly supports compliance with Republic Act No. 10173 (2012) and the National Privacy Commission (n.d.) guidelines for access-controlled processing of sensitive health information in government systems.

From a deployment perspective, implementing H-MAP as a path-based directory at hrmo.paranaque.gov.ph/hmap avoids the need for new DNS configuration, additional SSL certificate provisioning, or separate server allocation. The proposed system will share the existing server resources, PHP runtime, and MySQL instance of the HRMO portal, minimizing infrastructure requirements and simplifying both initial deployment and ongoing maintenance.

Dashboard and Data Visualization for Public Health

Data visualization dashboards are increasingly recognized as essential tools for translating complex surveillance data into actionable public health intelligence. Dowding et al. (2015) conducted a review of dashboard implementations for improving patient care and found that effective dashboards must present the most important surveillance indicators in a single, visually clear interface that enables immediate understanding and action. In public health surveillance contexts, their review emphasized that trend line charts and bar charts are most effective for communicating epidemiological time-series data to health practitioners, as these formats align with the temporal mental models that health personnel use when interpreting surveillance outputs.

Dong et al. (2020) demonstrated the operational effectiveness of a web-based epidemiological dashboard that combined real-time GIS mapping with tabular and chart-based case summaries during the COVID-19 pandemic. Their work illustrates that browser-based dashboards can serve large user bases across diverse technical environments when built on well-structured open-source technologies. Chart.js, an open-source JavaScript library for interactive data visualizations, provides the charting foundation for H-MAP. Research on AI-assisted surveillance dashboards by Srivastava et al. (2024) further establishes that embedding AI outputs within accessible visualization tools bridges the gap between machine learning model accuracy and practical utility for health decision-makers, which is a core design principle of H-MAP.

Role-Based Access Control and Data Privacy in Health Systems

Role-Based Access Control (RBAC) is a security model that restricts system access based on the roles assigned to individual users, ensuring that each user can only access functions and data appropriate to their responsibilities. In health information systems, RBAC is a foundational security requirement because such systems process sensitive personal information including patient identities, addresses, disease diagnoses, and outcome data. Republic Act No. 10173, the Data Privacy Act of 2012, requires that health information systems implement access controls that limit data exposure to personnel with a legitimate need for the information (Republic Act No. 10173, 2012). The National Privacy Commission (n.d.) has issued guidelines emphasizing that personal data collected for public health purposes must be protected through technical and organizational measures appropriate to the risk.

H-MAP will implement RBAC through a two-layer access control architecture. The first layer will be provided by the HRMO portal authentication, which verifies that the

accessing user is a currently active employee of the City Government of Parañaque before granting any access to H-MAP. The second layer will be implemented within H-MAP itself, assigning one of three role levels to each authenticated user: Data Encoder, Health Analyst, and Administrator. Patient-level data will be accessible only to Administrators with documented public health necessity, while dashboard visualizations and AI machine learning outputs will operate on aggregated, anonymized case counts, consistent with the data minimization principle under Republic Act No. 10173 (2012).

Software Quality Models and Evaluation Frameworks

ISO/IEC 25010 defines an internationally recognized software product quality model that specifies quality characteristics including functional suitability, reliability, usability, performance efficiency, security, maintainability, and portability (ISO, 2011). For applied developmental research in the Philippines, ISO/IEC 25010 is widely adopted as the conceptual basis for system evaluation because it provides a systematic and internationally validated framework for assessing software quality across multiple dimensions. The ISO (2011) standard is particularly important in public health systems, where reliability and accuracy are critical to the integrity of health decision-making. For H-MAP, five quality characteristics will serve as evaluation dimensions: Functionality, Reliability, Usability, Performance Efficiency, and Maintainability.

Foreign Studies

International research on web-based disease surveillance systems and AI-assisted outbreak detection has consistently demonstrated the effectiveness of machine learning and browser-rendered GIS tools in enhancing public health response. Boulos et al. (2011) found that open-source web mapping platforms provided sufficient functionality for disease mapping

tasks previously requiring proprietary desktop GIS software, with zero licensing costs identified as a key practical advantage for resource-constrained health agencies. Their review directly supports the Leaflet.js and OpenStreetMap approach adopted in H-MAP.

Dong et al. (2020) developed the Johns Hopkins University COVID-19 Dashboard, integrating real-time GIS mapping with case counts across multiple geographic scales. The system demonstrated that web-based spatial surveillance dashboards are operationally effective at scale and that open, browser-based architectures can serve diverse user bases across institutional and geographic boundaries, supporting the dashboard design approach of H-MAP. Shaman and Karspeck (2012) demonstrated successful influenza forecasting using time-series modeling, establishing that machine learning methods can generate reliable short-term outbreak predictions from historical surveillance data.

Chakraborty et al. (2019) applied Prophet to dengue forecasting in tropical urban settings and reported mean absolute percentage errors substantially lower than those achieved by traditional ARIMA models, validating Prophet's suitability for disease forecasting in contexts directly comparable to Parañaque City. Shi et al. (2016) validated ensemble machine learning for sub-city disease risk classification in Singapore, finding that Random Forest classifiers outperformed single decision tree and logistic regression approaches in predicting geographic units at high outbreak risk. Srivastava et al. (2024) provided a comprehensive review of AI roles in early detection of infectious diseases, confirming that AI-assisted surveillance systems improve detection speed and decision-support quality across a range of health system contexts.

Local Studies and Contextual Research

In the Philippine context, Carvajal et al. (2018) represents the most directly relevant local machine learning study. Using dengue incidence and meteorological data from Metropolitan Manila covering 2009 to 2013, they compared General Additive Modeling, SARIMA, Random Forest, and Gradient Boosting for dengue forecasting accuracy. Their study found that Random Forest achieved competitive predictive performance and revealed temporal patterns associated with meteorological variables that single-technique models missed. Critically, however, their study produced standalone forecasting models and did not integrate outputs into an operational surveillance platform accessible to city health office personnel. This limitation is explicitly addressed by H-MAP.

Olana et al. (2025), in the most current Philippine-specific study, compared SARIMA, Neural Network Autoregression, Random Forest, LSTM, TBATS, and Prophet for nationwide dengue forecasting using monthly data from 2017 to 2024. Their study confirmed the strong seasonality of Philippine dengue incidence and the viability of multiple machine learning approaches for time-series forecasting in the Philippine data environment. Like Carvajal et al. (2018), however, the study focused on model accuracy rather than operational deployment, leaving the gap of accessible, web-based AI surveillance platforms at the city level unaddressed. H-MAP responds directly to this gap.

Studies evaluating the PIDSR system at the local level have consistently identified the gap between national data collection and local analytical capability as a persistent limitation of city-level public health management (DOH, 2014). Local health offices possess years of surveillance data but lack the technical platforms to exploit that data for spatial visualization or AI-assisted forecasting. This finding directly motivates the development of H-MAP as a locally deployed analytical layer on top of existing PIDSR data.

Foreign Systems Related to H-MAP

Freifeld et al. (2008) developed HealthMap, a global disease outbreak monitoring platform that processes internet-based data sources to detect emerging disease threats through automated classification algorithms. While operating at a global scale fundamentally different from H-MAP's city-level scope, HealthMap demonstrates the operational feasibility of AI-driven disease detection as a complement to traditional surveillance reporting. The CDC Early Aberration Reporting System (EARS) provides a publicly documented model for threshold-based automated outbreak detection using moving averages and standard deviation methods (CDC, 2019), the same mathematical foundations that H-MAP will implement for its automated epidemic threshold alerting module. The Dong et al. (2020) COVID-19 Dashboard further demonstrates scalable architecture combining GIS, relational databases, and web technologies in a format that benchmarks H-MAP's integrated surveillance dashboard design.

Local Systems Related to H-MAP

The DOH PIDSR system and the Field Health Services and Information System (FHSIS) provide the national framework for disease data collection and aggregation in the Philippines (DOH, 2014). Both systems are oriented toward national-level reporting and aggregation rather than city-level spatial analysis and real-time alerting. Submission of data to these national systems does not automatically produce barangay-level heat maps, epidemic threshold alerts, or AI-generated forecasts for the submitting city health office. This structural limitation reinforces the need for supplementary, city-level analytical platforms such as H-MAP.

Several Metro Manila LGUs have developed or piloted internal disease monitoring dashboards using general-purpose business intelligence tools connected to spreadsheet data

sources. While these provide improved visualization over raw spreadsheets, they do not include geospatial heat mapping, automated threshold alerting, or AI machine learning forecasting. H-MAP offers a more complete analytical platform specifically designed for PIDS-R-formatted surveillance data with AI-assisted prediction capabilities that general-purpose tools do not natively provide.

Mapping of Related Literature to H-MAP System Features

The reviewed literature provides direct conceptual and empirical support for each of H-MAP's five functional modules. The HRMO Authentication Module is grounded in data privacy literature establishing RBAC and employment-verified access control as requirements for health information systems processing personal disease data under Republic Act No. 10173 (2012). The Data Digitization Module is supported by PIDS-R framework documentation establishing the appropriate data structure for city-level disease case management in the Philippines (DOH, 2014).

The Heat Map Visualization Module is supported by Lawson (2018), Boulos et al. (2011), and Nykiforuk and Flaman (2011), which collectively validate kernel density estimation and open-source web mapping as effective tools for communicating disease distribution to health practitioners. The Disease Monitoring and Alert Module is grounded in WHO (2018) EWARN methodology and CDC (2019) EARS documentation, which establish the two-standard-deviation epidemic threshold and moving average aberration detection as the standard methods for automated outbreak detection in tropical communicable disease surveillance. The AI Machine Learning Prediction Module draws on Taylor and Letham (2018), Chakraborty et al. (2019), and Olana et al. (2025) for Prophet forecasting validation, and on Breiman (2001), Carvajal et al. (2018), and Shi et al. (2016) for Random Forest

classification validation, with Srivastava et al. (2024) providing the broader AI in disease surveillance conceptual framework.

Synthesis of the Literature and Research Gap

The reviewed literature establishes several converging conclusions that collectively support the design and development of H-MAP. First, disease surveillance systems require both structured data collection and enhanced analytic capacity, and digital platforms significantly improve both dimensions compared to manual, spreadsheet-based approaches (WHO, 2018; CDC, 2019). Second, open-source web GIS technologies, particularly Leaflet.js with OpenStreetMap, have reached functional maturity sufficient for professional-grade disease surveillance mapping without proprietary licensing costs (Boulos et al., 2011; Nykiforuk & Flaman, 2011). Third, the WHO EWARN epidemic threshold methodology and CDC EARS provide statistically principled, operationally established standards for automated outbreak detection that are directly compatible with PIDSR-formatted data (WHO, 2018; CDC, 2019; DOH, 2014).

Fourth, AI machine learning models including Prophet and Random Forest have demonstrated reliable performance in communicable disease forecasting and spatial risk classification in the Philippine context specifically. Carvajal et al. (2018) demonstrated that Random Forest achieves competitive accuracy for dengue forecasting in Metropolitan Manila. Olana et al. (2025), in the most current Philippine study, confirmed that Prophet and Random Forest are among the viable models for national-scale dengue forecasting using PIDSR-formatted data spanning 2017 to 2024. These two studies provide the strongest available local validation for the AI components of H-MAP. Fifth, Srivastava et al. (2024) established the broader conceptual basis for AI-assisted disease surveillance, confirming that embedding AI

within operational health information systems improves detection speed and decision-support quality across diverse health contexts. Sixth, Republic Act No. 10173 (2012) and National Privacy Commission guidelines establish data privacy and RBAC as non-negotiable requirements for health information systems processing personal disease data.

Despite these converging foundations, the reviewed literature reveals a specific and addressable research gap. Carvajal et al. (2018) and Olana et al. (2025), the two most relevant Philippine machine learning studies, both produced standalone forecasting models focused on predictive accuracy without embedding their outputs into operational web-based surveillance platforms. Neither study addressed the practical challenge of making AI-generated disease forecasts accessible to non-technical health personnel at the city epidemiology unit level. Existing LGU dashboards in Metro Manila use general-purpose tools without GIS or machine learning capability, and national DOH systems do not provide city-level spatial visualization or AI-assisted forecasting. H-MAP addresses this specific intersection of gaps by developing an authenticated, role-based, web-based platform that embeds AI-generated Prophet forecasts and Random Forest risk classifications within an accessible surveillance dashboard designed specifically for CESU personnel using 14 years of Parañaque City PIDSR data. The system's deployment within the HRMO portal infrastructure further addresses the practical access control requirement absent from all reviewed systems.

Conceptual Framework of the Study

The present study adopts the Input-Process-Output (IPO) model as its conceptual framework. This model is appropriate for development-oriented research because it presents the logical relationship between the resources and requirements of the study, the development and evaluation activities to be undertaken, and the expected project outputs. For H-MAP, the

IPO model describes how disease surveillance data, government IT infrastructure, and open-source AI development technologies will be transformed through structured development activities into a functional web-based AI-assisted disease surveillance system that will subsequently be evaluated for software quality and operational acceptability.

Conceptual Model of the Study

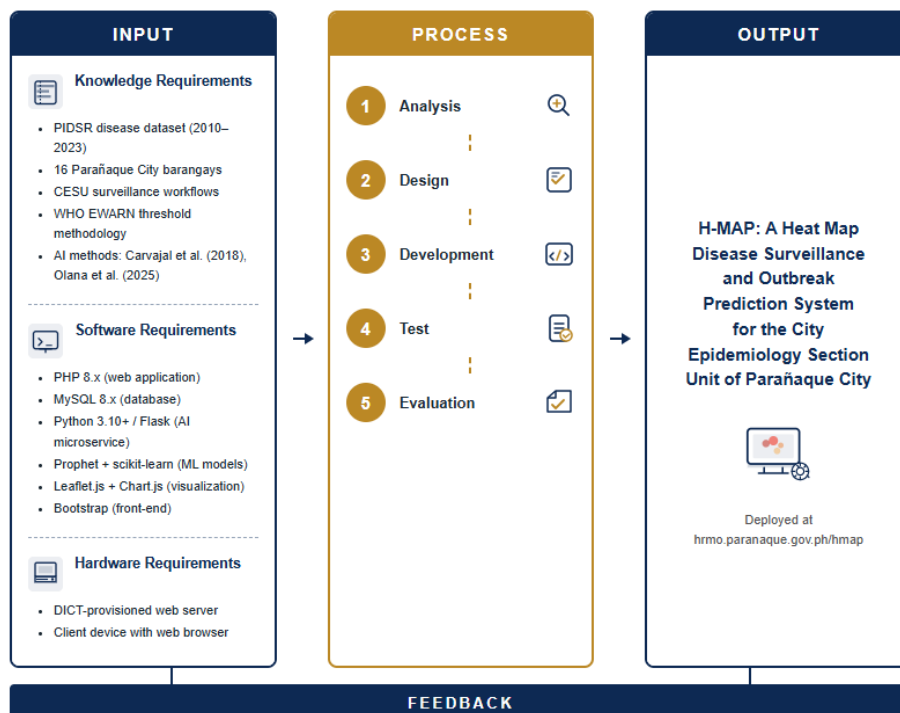


Figure 1. Conceptual Framework of the study

Input

Knowledge Requirements will include the PIDSR disease surveillance data from 2010 to 2023 provided by the Parañaque City CESU (over 30,000 records, seven priority diseases, 16 barangays), the PIDSR disease classification and reporting standards of the DOH (2014), the operational workflows and information needs of the City Epidemiology Section Unit, the WHO EWARN epidemic threshold methodology (WHO, 2018), and AI machine learning

methodologies validated in the Philippine context by Carvajal et al. (2018) and Olana et al. (2025).

Software Requirements will include PHP for server-side web application development, MySQL for relational data storage, Python 3 with Prophet, scikit-learn, and Flask for the AI machine learning microservice, Leaflet.js and the Leaflet.heat plugin for interactive web mapping, Chart.js for epidemiological data visualization, and the existing HRMO portal codebase and database for authentication integration.

Infrastructure Requirements will comprise the existing DICT-provisioned web server hosting hrmo.paranaque.gov.ph, the MySQL database instance already supporting the HRMO portal, and the PHP and Python runtime environments on the server.

Process

The Process stage will encompass the complete development and evaluation lifecycle of H-MAP: Requirements Analysis involving structured consultation with CESU staff and review of PIDSR data and workflows; System Design including architecture planning, database schema design, and AI machine learning model selection consistent with methodologies validated by Carvajal et al. (2018), Olana et al. (2025), Taylor and Letham (2018), and Breiman (2001); Development involving iterative construction of all five H-MAP modules and AI model training; Testing through functional, reliability, performance, and usability protocols; and Evaluation through ISO/IEC 25010-based questionnaire administration to CESU health officials and IT professionals (ISO, 2011).

Output

The primary output will be the H-MAP system, a PHP-based web application with five integrated modules deployed at hrmo.paranaque.gov.ph/hmap on the DICT-provisioned

government server. The secondary output will be the evaluation results documenting H-MAP's software quality based on ISO/IEC 25010 criteria (ISO, 2011), providing a formal basis for assessing the system's operational readiness and acceptability among its intended users.

Operational Definition of Terms

Artificial Intelligence (AI) Machine Learning Microservice. The Python Flask application to be deployed internally on the DICT server that implements the Prophet forecasting model and Random Forest risk classifier, serving AI-generated predictions to the PHP web application through an internal HTTP API, consistent with methodologies validated by Carvajal et al. (2018), Olana et al. (2025), and Taylor and Letham (2018).

Barangay Risk Classification. The output of the Random Forest machine learning model in H-MAP, categorizing each of Parañaque City's 16 barangays as Low, Moderate, or High disease risk for a selected week and disease, based on historical PIDSR case patterns and current trend features (Breiman, 2001; Carvajal et al., 2018).

Chart.js. An open-source JavaScript library for rendering interactive, browser-based data visualizations used in H-MAP to display epidemic curves, trend charts, and AI forecast outputs on the surveillance dashboard.

Data Digitization. The process of encoding disease case records from PIDSR-formatted health facility reports into the H-MAP MySQL database through the system's structured web-based data entry interface, consistent with DOH (2014) reporting standards.

Disease Surveillance. The systematic, ongoing collection, analysis, and communication of disease incidence data for the purpose of detecting outbreaks, monitoring trends, and supporting public health decision-making within Parañaque City (WHO, 2018).

Epidemic Threshold. The statistical threshold defined as the mean weekly case count plus two standard deviations based on five years of historical data for the same morbidity week, consistent with WHO EWARN methodology (WHO, 2018). Exceedance of this threshold in H-MAP will trigger an automated outbreak alert.

H-MAP. The Heat Map Disease Surveillance and Outbreak Prediction System proposed in the present study for the City Epidemiology Section Unit of Parañaque City, to be implemented as a PHP web application with AI machine learning capabilities deployed at hrmo.paranaque.gov.ph/hmap.

Heat Map. A spatial visualization layer rendered by Leaflet.js using kernel density estimation that represents the geographic density of disease cases as a color-coded intensity surface overlaid on the Parañaque City base map (Lawson, 2018).

HRMO Authentication. The access control mechanism of H-MAP that will require users to be authenticated through the Parañaque City HRMO portal and verified as currently active employees in the HRMO database before accessing any disease surveillance features, consistent with Republic Act No. 10173 (2012).

ISO/IEC 25010. An international standard defining a software product quality model with characteristics including functional suitability, reliability, usability, performance efficiency, and maintainability, used as the evaluation framework for H-MAP (ISO, 2011).

Leaflet.js. An open-source JavaScript library for interactive web mapping used in H-MAP to render the base map, heat map overlay, and barangay risk markers on the surveillance dashboard (Boulos et al., 2011).

PIDSR. The Philippine Integrated Disease Surveillance and Response system, the national framework that defines the disease categories, case classifications, reporting timelines, and data formats used in H-MAP's data management and alerting modules (DOH, 2014).

Prophet. An open-source time-series forecasting library developed by Meta, to be used in H-MAP's AI component to generate four-week-ahead weekly case count predictions for priority diseases based on 14 years of historical PIDSR data (Taylor & Letham, 2018), with forecasting performance validated in the Philippine dengue context by Olana et al. (2025).

Random Forest. An ensemble machine learning algorithm to be used in H-MAP's AI component for barangay risk classification, trained on historical PIDSR data to predict Low, Moderate, or High disease risk for each Parañaque City barangay for a given week and disease (Breiman, 2001), with performance validated in the Metro Manila dengue context by Carvajal et al. (2018).

Role-Based Access Control. The security model to be implemented in H-MAP that will assign one of three access levels, namely Data Encoder, Health Analyst, or Administrator, to each authenticated user, consistent with Republic Act No. 10173 (2012).

WHO EWARN. The World Health Organization Early Warning and Response Network methodology for epidemic threshold calculation, which defines the outbreak alert threshold as the mean plus two standard deviations of historical weekly case counts (WHO, 2018), the same method to be adopted in H-MAP's automated alerting module.

Chapter 3

METHODOLOGY

This chapter presents the research methodology to be employed in the design, development, testing, and evaluation of H-MAP. It describes the research design, the complete project development plan including system architecture, module specifications, AI machine learning model design, and database management, the instruments and testing procedures to be applied, the evaluation model and respondents, and the statistical treatment of evaluation data.

Project Design

This study will employ an Applied Developmental Research Design, in which the primary objective is the design, construction, testing, and formal evaluation of a functional software system that addresses a documented operational need within a specific institutional context. Applied developmental research is appropriate here because the contribution of the study is the H-MAP system itself, together with evidence of its quality based on ISO/IEC 25010 evaluation criteria (ISO, 2011). The study does not aim primarily to generate new theoretical knowledge but to produce and formally evaluate a practical artifact of direct operational value to the City Epidemiology Section Unit, specifically addressing the operational gap identified in the Philippine machine learning literature by Carvajal et al. (2018) and Olana et al. (2025).

Development will follow an Agile-Inspired Iterative Prototyping Model, with the project progressing through repeated cycles of planning, development, testing, and feedback-driven refinement within a one-month development timeline. This focused scope is intentional:

rather than pursuing feature breadth, H-MAP prioritizes the delivery of five well-tested modules that collectively address the core surveillance needs of the CESU. The iterative approach is appropriate because the HRMO authentication integration, AI machine learning microservice communication, and heat map parameter calibration each require short cycles of testing and adjustment that benefit from incremental rather than linear development.

The complete development lifecycle will consist of the following five phases:

- a. Requirements Analysis: structured consultation with CESU health officials to identify surveillance workflows, information needs, priority diseases, and system requirements; review of historical PIDSR data to confirm AI machine learning model feasibility.
- b. System Design: database schema design, system architecture planning, module interface design, and AI machine learning model selection consistent with Carvajal et al. (2018), Olana et al. (2025), Taylor and Letham (2018), and Breiman (2001).
- c. Development: iterative construction of all five H-MAP modules and AI model training within the one-month development window.
- d. Testing: functional, reliability, performance, and usability testing using documented test cases across all modules.
- e. Evaluation: ISO/IEC 25010-based questionnaire administration to CESU health officials and IT professionals (ISO, 2011).

System Overview

H-MAP will be a PHP-based web application with an integrated AI Python machine learning microservice, to be deployed at hrmo.paranaque.gov.ph/hmap on the DICT-

provisioned server that hosts the Parañaque City HRMO portal. The proposed system will provide authorized city government employees with an integrated disease surveillance dashboard featuring interactive heat maps, automated outbreak alerts, trend visualizations, and AI machine learning-based forecasts derived from Parañaque City's 14-year PIDSR disease case dataset. The system is designed to be completable within a one-month development period by focusing on five clearly scoped modules, each addressing a specific and documented operational need of the CESU.

The proposed system will enable authorized health officials and data personnel to:

- a. Log in using their existing HRMO portal credentials, with access granted only to currently active employees, consistent with Republic Act No. 10173 (2012).
- b. Encode new disease case records using a structured data entry form aligned with PIDSR reporting standards (DOH, 2014).
- c. View an interactive Leaflet.js heat map displaying current geographic distribution of disease cases across Parañaque City's 16 barangays, with barangay-level AI risk overlays derived from the Random Forest classifier (Breiman, 2001; Carvajal et al., 2018).
- d. Monitor real-time disease trends through Chart.js epidemic curve charts with automated color-coded epidemic threshold alerts consistent with WHO EWARN methodology (WHO, 2018).
- e. Access four-week-ahead AI forecasts for priority diseases using the Prophet model, presented as predicted case counts with confidence intervals (Taylor & Letham, 2018; Olana et al., 2025).

- f. Export summary surveillance reports for administrative reporting and DOH submission.

HRMO Portal Authentication Integration

H-MAP will use the HRMO portal's existing PHP session infrastructure for authentication. When an employee accesses `hrmo.paranaque.gov.ph/hmap`, the H-MAP PHP entry point will check for the presence of an active HRMO session. If no active session is found, the user will be redirected to the HRMO portal login page with the H-MAP path set as the post-login destination. If an active session is found, H-MAP will perform a secondary verification query against the HRMO MySQL database to confirm that the employee's current status is ACTIVE, ensuring that employees whose separation has been recorded by HR are denied access without requiring manual deactivation in a separate user management system. This design is consistent with access control principles required under Republic Act No. 10173 (2012) and the National Privacy Commission (n.d.) guidelines.

Following successful authentication and status verification, H-MAP will retrieve the employee's role assignment from a dedicated role mapping table. The three role levels to be implemented are:

- a. Data Encoder: access to case data entry and personal data viewing only, with no access to AI machine learning predictions, bulk exports, or system configuration.
- b. Health Analyst: full access to heat map visualization, epidemic threshold alerts, trend charts, AI Prophet forecasts, and Random Forest risk classifications, with no access to system configuration or user role management.
- c. Administrator: full access to all features including user role assignment, disease threshold configuration, AI model retraining triggers, and complete data export.

System Modules

H-MAP will be composed of five functional modules, scoped for completion within the one-month development timeline:

Module 1: Data Digitization and Records Management. This module will provide a structured web-based interface for encoding PIDSR disease case records into the H-MAP MySQL database. Input fields will mirror the PIDSR case report form, consistent with DOH (2014) reporting standards, including disease name, case classification, date of onset, date of admission, patient barangay, reporting health facility, age, sex, vaccination history, and case outcome. Input validation will check completeness and classification consistency before record insertion. A batch import function will support CSV upload of multiple records using a PIDSR-compatible template, facilitating transfer of historical data from existing Excel-based surveillance records.

Module 2: Heat Map Visualization. This module will render an interactive disease surveillance map using Leaflet.js with OpenStreetMap tile layers. Upon selecting a disease and date range, the module will retrieve geocoded case records from MySQL and pass them to the Leaflet.heat plugin, which will compute and render the KDE heat map as a color gradient overlay on the base map (Lawson, 2018). A GeoJSON barangay boundary layer will be rendered as a semi-transparent polygon overlay, with each barangay polygon colored Low (green), Moderate (yellow), or High (red) based on the current output of the AI Random Forest risk classifier retrieved from the Python machine learning microservice (Breiman, 2001; Carvajal et al., 2018).

Module 3: Disease Trend Monitoring and Epidemic Threshold Alert. This module will provide the core epidemiological surveillance dashboard. For the selected disease, the module

will query the H-MAP MySQL database to retrieve weekly case counts for the current year and compare them against pre-computed threshold values. Threshold values will be computed as the historical mean plus two standard deviations across the preceding five years, consistent with WHO EWARN methodology (WHO, 2018) and CDC EARS (CDC, 2019). Results will be displayed as a Chart.js line chart overlaying the current year epidemic curve against the historical mean and threshold. A color-coded status indicator will update automatically each week: green if current cases are below the historical mean, yellow if above the mean but below the threshold, and red if the threshold is exceeded.

Module 4: AI Machine Learning Prediction. This module will provide H-MAP's forward-looking AI-assisted analytical capabilities by communicating with the internal Python Flask microservice. When a Health Analyst or Administrator opens the prediction panel, the PHP application will send an HTTP request to the Flask service and await a JSON response containing: the Prophet four-week forecast with predicted case counts and confidence intervals (Taylor & Letham, 2018; Olana et al., 2025); the epidemic threshold status for each forecast week (WHO, 2018); and the Random Forest risk classification for each of the 16 barangays (Breiman, 2001; Carvajal et al., 2018), displayed as a colored polygon overlay on the Leaflet.js map. A visible note on the prediction panel will state that AI machine learning outputs are decision-support indicators and that epidemiological judgment of qualified health personnel is required for public health action decisions, consistent with the decision-support principle articulated by Srivastava et al. (2024).

Module 5: Reports and Data Export. This module will allow authorized users to generate and download summary surveillance reports. Data Encoders may export a personal data entry log. Health Analysts may generate disease summary reports including epidemic

curve charts and barangay case count tables. Administrators may export the complete case registry as a CSV file for DOH submission or archival purposes. All export actions will be recorded in the system audit trail, consistent with Republic Act No. 10173 (2012).

AI Python Machine Learning Microservice

The AI machine learning component of H-MAP will run as a Flask application on the same DICT server, listening on internal port 5001 and accessible only through localhost, not exposed to the public internet. The microservice will provide two endpoints: /predict/forecast for Prophet model execution, and /predict/risk for Random Forest classification. Both models will be pre-trained and loaded at startup for fast response without retraining on each request.

Prophet Model. One Prophet model will be trained per priority disease using the complete aggregated weekly case time-series from 2010 to 2023 (Taylor & Letham, 2018). The training process follows the approach validated by Olana et al. (2025) for Philippine dengue data, aggregating case records by disease and morbidity week into the two-column dataframe format required by Prophet (ds for week start date, y for case count). Each trained model will be serialized using Python's pickle module and stored on the server. Administrators may trigger model retraining through the H-MAP Administrator panel as new surveillance data is added.

Random Forest Model. The Random Forest classifier will be trained using labeled historical observations derived from the PIDSR dataset at the disease-barangay-week level, following the approach employed by Carvajal et al. (2018) for Metro Manila dengue data (Breiman, 2001). Each observation will be labeled High risk if the barangay's case count exceeded the epidemic threshold for that week, Moderate if between the historical mean and threshold, and Low if at or below the mean. The feature vector will include barangay identifier,

morbidity week number, calendar month, current week case count, same-week-prior-year case count, five-year historical mean, ratio of current to historical mean, and cumulative year-to-date case count for the barangay. A scikit-learn RandomForestClassifier with 200 estimators and class-weight balancing will be used.

Model Validation. Both models will be validated using a temporal hold-out approach consistent with Olana et al. (2025): training on 2010 to 2020 data and evaluating against 2021 to 2023 data. For the Prophet model, Mean Absolute Percentage Error will be reported for forecast weeks. For the Random Forest classifier, accuracy, precision, recall, and F1 score will be reported per risk class. Validation results will be documented in the system evaluation findings.

Development Environment

Server-Side Technologies:

- a. PHP 8.x: primary web application language, consistent with the existing HRMO portal.
- b. MySQL 8.x: relational database management system for case records, user roles, threshold tables, and system logs.
- c. Python 3.10 and higher: runtime environment for the AI Flask microservice, Prophet models, and scikit-learn classifiers.
- d. Flask 3.x: lightweight Python web framework serving the internal AI prediction API.
- e. Prophet: open-source time-series forecasting library validated for Philippine dengue data by Olana et al. (2025) and Taylor and Letham (2018).

- f. scikit-learn: machine learning library providing the RandomForestClassifier validated in the Metro Manila dengue context by Carvajal et al. (2018) and Breiman (2001).
- g. pandas and numpy: Python data processing libraries used in AI machine learning data preparation and feature engineering.

Client-Side Technologies:

- a. Leaflet.js: open-source JavaScript mapping library for interactive base map and heat map rendering (Boulos et al., 2011).
- b. Leaflet.heat plugin: Leaflet extension for kernel density estimation heat map generation (Lawson, 2018).
- c. Chart.js: open-source JavaScript charting library for epidemic curves, trend charts, and AI forecast visualizations.
- d. Bootstrap: responsive CSS framework for dashboard layout and interface components.

Hardware Requirements (Minimum):

- a. Server: Intel Core i5 or equivalent, 8 GB RAM, 50 GB available storage, Ubuntu Server 20.04 LTS or equivalent.
- b. Client: any device with a modern web browser; no additional software installation required.

System Architecture

H-MAP will follow a three-layer web application architecture with an additional AI machine learning service layer:

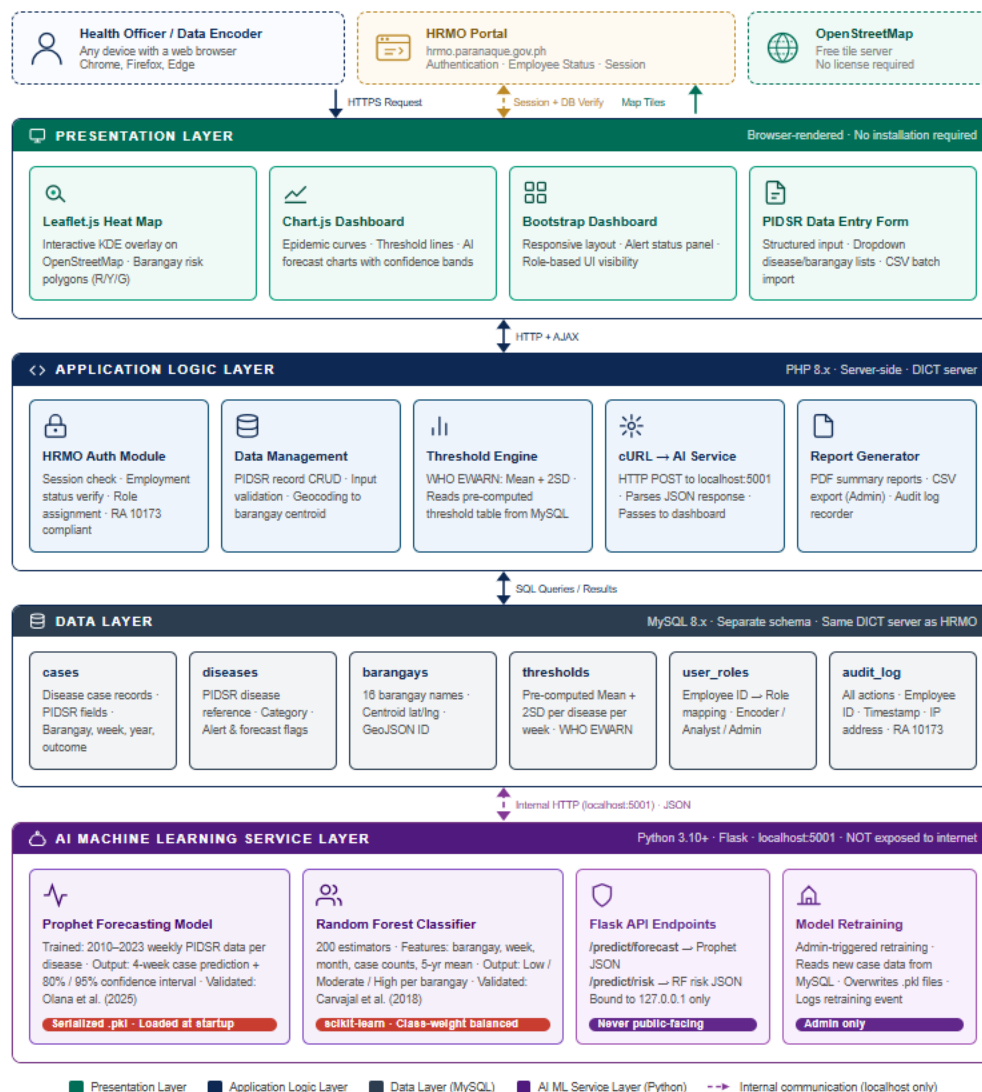


Figure 2. System Architecture

Presentation Layer. The browser-rendered interface composed of Leaflet.js map panels, Chart.js chart panels, Bootstrap-styled dashboard components, and PHP-generated HTML. All user interactions are handled at this layer, communicating with the application layer through HTTP form submissions and AJAX requests.

Application Logic Layer. The PHP codebase handling HRMO session and MySQL authentication verification, data validation and sanitization, MySQL queries for surveillance data retrieval and aggregation, epidemic threshold computation against stored threshold tables,

report generation, and internal cURL calls to the AI Python Flask microservice on localhost:5001.

Data Layer. The MySQL database containing the H-MAP case records table, barangay reference table with centroid coordinates, PIDSR disease reference table consistent with DOH (2014), pre-computed weekly epidemic threshold table based on WHO EWARN methodology (WHO, 2018), user role assignment table, and system audit log table. The database will share the same MySQL server instance as the HRMO portal but will use a separate schema to maintain clear data separation.

AI Machine Learning Service Layer. The Python Flask microservice running internally on localhost:5001, separate from the PHP application, maintaining pre-trained Prophet and Random Forest model objects in memory for fast AI prediction response. Methodology follows Carvajal et al. (2018), Olana et al. (2025), Taylor and Letham (2018), and Breiman (2001).

Operational Flow

- a. Employee accesses hrmo.paranaque.gov.ph/hmap through a web browser.
- b. PHP checks for active HRMO session; if absent, redirects to HRMO portal login.
- c. Upon confirmed active session, PHP queries HRMO MySQL to verify ACTIVE employment status, consistent with Republic Act No. 10173 (2012).
- d. H-MAP dashboard loads with default view showing the current week's heat map and epidemic curve.
- e. For heat map: PHP retrieves geocoded case records from MySQL; Leaflet.heat renders KDE overlay (Lawson, 2018); PHP sends cURL request to Flask /predict/risk; barangay AI risk polygons update on map.

- f. For trend chart: PHP retrieves weekly case counts and pre-computed WHO EWARN thresholds from MySQL (WHO, 2018); Chart.js renders epidemic curve with threshold line; color-coded status indicator updates.
- g. For AI forecast: PHP sends cURL request to Flask /predict/forecast; Prophet prediction JSON returns (Taylor & Letham, 2018; Olana et al., 2025); Chart.js renders forecast chart with confidence bands.
- h. User may generate and download a summary report from the Reports module.
- i. All actions are recorded in the system audit log, consistent with Republic Act No. 10173 (2012).

Database Design

The H-MAP MySQL database will use the following primary tables:

cases: individual PIDSR-aligned disease case records including case identifier, disease reference, case classification, date of onset, date of admission, barangay identifier, facility identifier, age, sex, outcome, morbidity week, morbidity month, year, entered-by employee identifier, and entry timestamp, consistent with DOH (2014) reporting fields.

TABLE: cases — Disease Case Records (PIDSR-aligned)				
Column Name	Data Type	Key		Description
case_id	INT UNSIGNED	PK	NN	Auto-increment unique case identifier
disease_id	INT UNSIGNED	FK	NN	References diseases.disease_id — PIDSR classification
case_classification	ENUM	NN		PIDSR values: 'Suspect', 'Probable', 'Confirmed', 'Negative'
date_onset	DATE	NN		Date of symptom onset (PIDSR required field)
date_admitted	DATE			Date of hospital/facility admission (nullable)
date_reported	DATE	NN		Date case was reported to CESU
barangay_id	INT UNSIGNED	FK	NN	References barangays.barangay_id — patient residence
facility_id	INT UNSIGNED	FK		References facilities.facility_id — reporting health center
age	TINYINT UNSIGNED			Patient age in years at time of onset
age_group	ENUM			'0-4', '5-9', '10-14', '15-19', '20-59', '60+
sex	ENUM			'Male', 'Female'
outcome	ENUM			'Alive', 'Died', 'Unknown'
morbidity_week	TINYINT UNSIGNED	NN	IDX	ISO epidemiological week number (1-53) — used for threshold computation
morbidity_month	TINYINT UNSIGNED	NN		Calendar month (1-12)
morbidity_year	YEAR	NN	IDX	Calendar year — used for historical aggregation
entered_by	INT UNSIGNED	FK	NN	References HRMO.employee_id — who encoded the record
entered_at	DATETIME	NN		Timestamp of record creation — DEFAULT CURRENT_TIMESTAMP
updated_at	DATETIME			Timestamp of last update — ON UPDATE CURRENT_TIMESTAMP
status_flag	ENUM	NN		'Active', 'Deleted' — soft delete for audit compliance (RA 10173)

Figure 3. Core Transaction Table: Cases

diseases: controlled PIDS disease reference list consistent with DOH (2014) classifications, including disease identifier, disease name, disease category, alert-enabled flag, and forecast-enabled flag.

TABLE: diseases			
Column	Type	Key	Description
disease_id	INT UNSIGNED	PK NN	Auto-increment disease identifier
disease_code	VARCHAR(20)	NN IDX	PIDS disease code (e.g., 'DEN', 'MEA', 'LEP')
disease_name	VARCHAR(100)	NN	Full disease name per PIDS list
disease_category	ENUM	NN	'Category 1' (immediate) or 'Category 2' (weekly)
alert_enabled	TINYINT(1)	NN	1 = threshold alerting active for this disease
forecast_enabled	TINYINT(1)	NN	1 = Prophet + RF models available for this disease
display_order	TINYINT UNSIGNED		Sort order in dropdown lists

Figure 4. Reference Table: diseases

barangays: Parañaque City barangay reference including barangay identifier, barangay name, population, centroid latitude, centroid longitude, and GeoJSON identifier for boundary layer matching.

TABLE: barangays			
Column	Type	Key	Description
barangay_id	INT UNSIGNED	PK NN	Auto-increment barangay identifier
barangay_name	VARCHAR(80)	NN	Official barangay name (16 Parañaque barangays)
population	INT UNSIGNED		Latest PSA population count
centroid_lat	DECIMAL(9, 6)	NN	Latitude of barangay centroid — used for KDE geocoding
centroid_lng	DECIMAL(9, 6)	NN	Longitude of barangay centroid — used for KDE geocoding
geojson_id	VARCHAR(40)		Feature ID in barangay boundary GeoJSON file
district	TINYINT UNSIGNED		Parañaque City district (1 or 2)

Figure 5. Reference Table: barangays

thresholds: pre-computed weekly epidemic thresholds for all monitored diseases, including disease identifier, morbidity week, five-year mean case count, standard deviation, and threshold value computed as mean plus two standard deviations consistent with WHO EWARN (WHO, 2018).

Column	Type	Key	Description
threshold_id	INT UNSIGNED	PK NN	Auto-increment threshold record identifier
disease_id	INT UNSIGNED	FK NN	References diseases.disease_id
morbidity_week	TINYINT UNSIGNED	NN	Epidemiological week (1–53)
baseline_years	VARCHAR(30)		Years used in computation e.g. '2018,2019,2020,2021,2022'
mean_cases	DECIMAL(8,4)	NN	5-year average case count for this disease-week
std_dev	DECIMAL(8,4)	NN	Standard deviation of baseline case counts
threshold_value	DECIMAL(8,4)	NN	Mean + 2×SD — WHO EWARN epidemic alert threshold
last_updated	DATETIME	NN	Timestamp of last threshold recomputation

Figure 6. Reference Table: thresholds

facilities: stores the reference data for all health centers and sentinel sites within Parañaque City that report disease cases to the CESU. Its fields include the facility identifier, official facility name, facility type of either Health Center, Sentinel Site, or Hospital, the barangay identifier indicating the facility location, a sentinel flag indicating PIDSIR sentinel reporting status, and the facility GPS latitude and longitude for map pin display on the surveillance dashboard.

Column	Type	Key	Description
facility_id	INT UNSIGNED	PK NN	Auto-increment facility identifier
facility_name	VARCHAR(120)	NN	Official health center or sentinel site name
facility_type	ENUM	NN	Health Center/ 'Sentinel Site', 'Hospital'
barangay_id	INT UNSIGNED	FK	References barangays.barangay_id — location of the facility
is_sentinel	TINYINT(1)		1 = designated PIDSIR sentinel reporting site for the CESU
facility_lat	DECIMAL(9,6)		Facility GPS latitude for map pin display
facility_lng	DECIMAL(9,6)		Facility GPS longitude for map pin display

Figure 7. Reference Table: facilities

user_roles: H-MAP role assignments for authenticated HRMO employees including employee identifier, role designation, assigning administrator, and assignment timestamp.

Column	Type	Key	Description
role_id	INT UNSIGNED	PK NN	Auto-increment role record
employee_id	INT UNSIGNED	NN IDX	HRMO portal employee_id (external reference)
role	ENUM	NN	'Encoder', 'Analyst', 'Administrator'
assigned_by	INT UNSIGNED	FK	employee_id of assigning Administrator
assigned_at	DATETIME	NN	Role assignment timestamp
is_active	TINYINT(1)	NN	1 = active role, 0 = revoked

Figure 8. Access Control, AI Output Cache & Audit Tables: user_roles

audit_log: records all user actions including employee identifier, action type, target record, timestamp, and IP address, consistent with accountability requirements under Republic Act No. 10173 (2012).

Column	Data Type	Key	Description
log_id	BIGINT UNSIGNED	PK NN	Auto-increment log entry identifier — BIGINT used to accommodate long-term high-volume audit records
employee_id	INT UNSIGNED	NN IDX	HRMO employee_id of the user who performed the action — external reference to HRMO employee table
action_type	VARCHAR(50)	NN	Type of action performed: 'INSERT', 'UPDATE', 'DELETE', 'LOGIN', 'LOGOUT', 'EXPORT', 'AI_REQUEST', 'RETRAIN'
target_table	VARCHAR(40)		Name of the database table that was affected by the action (e.g., 'cases', 'user_roles')
target_id	INT UNSIGNED		Primary key value of the specific record that was affected by the action
old_values	JSON		JSON snapshot of the record's field values before UPDATE or DELETE — enables full before-state reconstruction
new_values	JSON		JSON snapshot of the record's field values after INSERT or UPDATE — enables full after-state reconstruction
ip_address	VARCHAR(45)		Client IP address of the request — VARCHAR(45) supports both IPv4 and IPv6 formats
action_at	DATETIME	NN IDX	Exact timestamp of the logged action — DEFAULT CURRENT_TIMESTAMP; indexed for fast date-range audit queries

Figure 9. Access Control, AI Output Cache & Audit Tables: audit_log

ai_predictions: stores the output log of all AI machine learning predictions generated by the Python Flask microservice. It records both Prophet time-series forecast results and Random Forest barangay risk classification results in the same table, distinguished by the **prediction_type** field. For Prophet records, the table stores the predicted case count and 80-percent confidence interval bounds. For Random Forest records, it stores the barangay identifier and the risk level output of Low, Moderate, or High. The table maintains a complete historical log of all AI predictions requested by authorized Health Analysts and Administrators, supporting both trend review and accountability documentation.

TABLE: ai_predictions — AI Model Output Log (Prophet Forecasts + Random Forest Risk Classifications)			
Column	Data Type	Key	Description
prediction_id	INT UNSIGNED	PK NN	Auto-increment AI prediction log identifier
disease_id	INT UNSIGNED	FK NN	References diseases.disease_id — disease for which the prediction was generated
prediction_type	ENUM	NN	'Prophet_Forecast' for time-series predictions; 'RF_Risk' for Random Forest barangay risk classifications
generated_at	DATETIME	NN	Timestamp when the Python Flask microservice generated this prediction result
forecast_week_start	DATE		Start date of the 4-week forecast window (Prophet only; NULL for RF_Risk records)
predicted_cases	DECIMAL(5, 2)		Prophet predicted case count for the forecast week (yhat value); NULL for RF_Risk records
confidence_lower	DECIMAL(5, 2)		80% lower confidence interval bound (yhat_lower from Prophet); NULL for RF_Risk records
confidence_upper	DECIMAL(5, 2)		80% upper confidence interval bound (yhat_upper from Prophet); NULL for RF_Risk records
barangay_id	INT UNSIGNED	FK	References barangays.barangay_id — the barangay being classified (RF_Risk only; NULL for Prophet records)
risk_level	ENUM		Random Forest output classification: 'Low', 'Moderate', or 'High' risk level for the barangay-week
requested_by	INT UNSIGNED		HRMO employee_id of the Analyst or Administrator who triggered the AI prediction request

Figure 10. Access Control, AI Output Cache & Audit Tables: ai_predictions

The relationships among the H-MAP database tables are governed by foreign key constraints implemented at the MySQL InnoDB engine level. The cases table is the central transaction table and holds foreign keys referencing the **disease_id** of the diseases table, the **barangay_id** of the barangays table, and the **facility_id** of the facilities table, establishing one-to-many relationships in each instance. The facilities table additionally holds a foreign key referencing the barangays table, reflecting that each facility is located within one barangay. The thresholds table references the diseases table through a foreign key with a CASCADE delete rule, ensuring threshold records are automatically removed when a disease record is

deleted, and enforces a unique constraint on the combination of `disease_id` and `morbidity_week` to guarantee one threshold value per disease per epidemiological week. The `ai_predictions` table holds foreign keys to both the `diseases` table and the `barangays` table, with the barangay reference accepting NULL values to accommodate Prophet forecast records that are not barangay-specific. The `user_roles` table references the HRMO employee database through a logical cross-schema reference rather than an enforced foreign key constraint, as the two tables reside in separate database schemas — H-MAP's `hmap_db` and the HRMO portal's own database. Similarly, the `entered_by` field in the `cases` table and the `employee_id` field in the `audit_log` table reference the HRMO employee identifier through logical cross-schema references, with employee identity verified through the HRMO session at the point of data entry rather than through a database-level constraint.

RELATIONSHIPS & FOREIGN KEY CONSTRAINTS				
Parent Table	→	Child Table	Cardinality	Constraint and Notes
<code>diseases.disease_id</code>	→	<code>cases.disease_id</code>	1 : MANY	ON DELETE RESTRICT — one disease has many case records; prevents deletion of referenced disease
<code>barangays.barangay_id</code>	→	<code>cases.barangay_id</code>	1 : MANY	ON DELETE RESTRICT — one barangay has many case records; geocoding anchor for heat map
<code>facilities.facility_id</code>	→	<code>cases.facility_id</code>	1 : MANY	ON DELETE SET NULL — case record is preserved if facility is removed from the reference list
<code>barangays.barangay_id</code>	→	<code>facilities.barangay_id</code>	1 : MANY	ON DELETE SET NULL — one barangay may have multiple health centers or sentinel sites
<code>diseases.disease_id</code>	→	<code>thresholds.disease_id</code>	1 : MANY	ON DELETE CASCADE — threshold records are deleted with their parent disease; UNIQUE (<code>disease_id</code> , <code>morbidity_week</code>)
<code>diseases.disease_id</code>	→	<code>ai_predictions.disease_id</code>	1 : MANY	ON DELETE RESTRICT — prediction history is preserved for audit and trend review purposes
<code>barangays.barangay_id</code>	→	<code>ai_predictions.barangay_id</code>	1 : MANY	ON DELETE SET NULL — RF risk prediction records per barangay per week; NULL for Prophet records
HRMO DB. <code>employee_id</code>	→	<code>cases.entered_by</code>	1 : MANY	Cross-schema logical reference — no FK constraint enforced; employee identity verified via HRMO session at data entry
HRMO DB. <code>employee_id</code>	→	<code>user_roles.employee_id</code>	1 : 1	Cross-schema logical reference — UNIQUE (<code>employee_id</code>) enforces one H-MAP role per HRMO employee
HRMO DB. <code>employee_id</code>	→	<code>audit_log.employee_id</code>	1 : MANY	Cross-schema logical reference — all user actions are logged per employee for RA 10173 accountability

PK: Primary Key FK: Foreign Key NN: NOT NULL IDX: Indexed Total tables: 7 | Schema: hmap_db | Engine: InnoDB | Charset: utf8mb4

Figure 11. Entity Relationship and Foreign Key Constraints

Software Testing Plan

A multi-level testing strategy will be applied across all H-MAP modules prior to formal evaluation. Testing will be organized as follows:

A. **Functionality Testing.** Functional test cases will be developed for each module and major system operation, specifying the module, operational procedure, expected result, and observed system response. Test cases will cover HRMO session detection and redirect behavior; employment status verification; role-based access control enforcement; data entry validation and database insertion; heat map rendering for different disease and date range inputs; epidemic curve chart accuracy against WHO EWARN thresholds (WHO, 2018); Flask AI microservice communication from PHP; Prophet forecast JSON parsing and chart rendering (Taylor & Letham, 2018; Olana et al., 2025); and Random Forest risk classification map overlay accuracy (Breiman, 2001; Carvajal et al., 2018).

B. **Reliability Testing.** Reliability testing will evaluate H-MAP's consistent performance under repeated use, including repeated execution of the full dashboard load sequence across multiple sessions and user role levels, and testing of AI machine learning prediction request handling under repeated calls. The system's behavior when the Flask microservice is temporarily unavailable will be verified to ensure that the PHP dashboard displays an informative message without failing.

C. **Performance Testing.** Performance testing will measure response times for key operations: initial dashboard load time, heat map rendering time for datasets of varying case volume, Chart.js chart rendering time, and Flask AI machine learning prediction response time from PHP cURL call to JSON receipt. Target response times are set at under 3 seconds for dashboard load and under 5 seconds for AI prediction response, consistent with web application usability standards.

D. **Usability Testing.** Usability testing will be conducted with selected CESU health official respondents who will perform standardized task sequences including logging in

through the HRMO portal, encoding a disease case record, navigating the heat map, interpreting the epidemic threshold alert, and accessing an AI disease forecast. Post-task questionnaire items will assess ease of navigation, clarity of map and chart visualizations, understandability of AI prediction outputs, and overall interface satisfaction, following ISO 9241 principles.

Evaluation Model

The formal evaluation of H-MAP will use selected characteristics of the ISO/IEC 25010 software product quality model (ISO, 2011) and an adopted software evaluation instrument. The selected quality characteristics are:

- a. **Functionality:** the degree to which H-MAP provides all intended disease surveillance, heat mapping, epidemic threshold alerting, and AI machine learning prediction features.
- b. **Reliability:** the degree to which H-MAP, including the PHP web application and the Python Flask AI microservice, performs its functions consistently under normal operational conditions.
- c. **Usability:** the degree to which CESU health officials and data encoders can use H-MAP effectively and with satisfaction to accomplish surveillance tasks, assessed following ISO 9241 principles.
- d. **Performance Efficiency:** the degree to which H-MAP renders maps, charts, and AI machine learning predictions within acceptable response times under the DICT server's resource constraints.

- e. Maintainability: the degree to which the PHP codebase and Python AI machine learning models can be modified and updated to accommodate new diseases or changed surveillance requirements (ISO, 2011).

Respondents will be purposively selected from:

- a. 25 to 30 health officials, medical officers, epidemiologists, and data personnel from the City Epidemiology Section Unit and affiliated health centers and sentinel sites of Parañaque City.
- b. Five Information Technology professionals with expertise in web systems, government IT, or health informatics.

The evaluation instrument will be a structured 5-point Likert-scale questionnaire adapted from ISO/IEC 25010 criteria (ISO, 2011), rated as follows:

- a. 5: Outstanding / Strongly Agree
- b. 4: Very Acceptable / Agree
- c. 3: Acceptable / Neutral
- d. 2: Needs Improvement / Disagree
- e. 1: Poor / Strongly Disagree

Data Processing and Statistical Treatment

The following descriptive statistical methods will be applied to evaluation response data:

- a. Frequency Distribution: summarizing the distribution of respondent ratings across the five-point scale for each evaluation item.

- b. Weighted Mean: computed as the sum of (rating x frequency) divided by total respondents for each item and each ISO/IEC 25010 quality characteristic, providing a single numerical indicator of overall acceptability (ISO, 2011).
- c. Standard Deviation: computed per item to measure variability in respondent ratings and indicate consensus or divergence of assessment.

Weighted mean scores will be interpreted against the following acceptability scale:

- a. 4.21 to 5.00: Outstanding / Very Acceptable
- b. 3.41 to 4.20: Acceptable / Agreeable
- c. 2.61 to 3.40: Moderately Acceptable
- d. 1.81 to 2.60: Needs Improvement
- e. 1.00 to 1.80: Poor / Unacceptable

Acceptance Criterion: H-MAP will be considered to have met the evaluation standard if the overall weighted mean across all five ISO/IEC 25010 quality characteristics is at least 3.41. Inferential statistical analysis will not be conducted given the descriptive intent of the evaluation phase and the sample size consistent with purposive sampling in applied developmental research.

Security Considerations

H-MAP's security architecture addresses the sensitivity of disease surveillance data under Republic Act No. 10173, the Data Privacy Act of 2012 (Republic Act No. 10173, 2012), and the applicable guidelines of the National Privacy Commission (n.d.). The primary security mechanisms to be implemented are:

- a. HRMO-Based Access Control: access restricted to active Parañaque City government employees through HRMO portal session verification and real-time employment status checking, with no public access permitted.
- b. Role-Based Feature Restriction: RBAC ensuring Data Encoders cannot access AI predictions or bulk exports; Health Analysts cannot access system configuration; and all Administrator actions are logged, consistent with Republic Act No. 10173 (2012).
- c. Internal AI Service Isolation: the Python Flask AI microservice bound to localhost:5001, accessible only from the PHP application on the same server, not exposed to any external network interface.
- d. Input Sanitization and Injection Prevention: all user-submitted data validated and sanitized using PHP prepared statements with parameterized queries via PDO.
- e. Audit Logging: all data entry, modification, deletion, export, and AI prediction access events recorded in the audit log with employee identifier, action type, timestamp, and IP address, consistent with Republic Act No. 10173 (2012).
- f. Data Minimization in Display: dashboard visualizations and AI machine learning outputs will operate on aggregated, barangay-level case counts without exposing individual patient identifiers, consistent with the data minimization principle under Republic Act No. 10173 (2012) and National Privacy Commission (n.d.) guidelines.

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